

RoMathExam: A Longitudinal Dataset of Romanian Math Exams (1895–2025) with a Seven-Decade Core (1957–2025)

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Abstract. AI in Education research increasingly relies on authentic, curriculum-grounded assessment data, yet large, well-structured exam corpora remain scarce for many languages and educational systems. We introduce RoMathExam, a longitudinal dataset of Romanian high-school mathematics exams spanning 1895–2025, with a robust standardized core for 1957–2025. The dataset contains 10,592 mathematics problems organized into 600+ complete exam sets across multiple tracks (M1–M4), covering both official national examination sessions and ministry-published training variants.

Beyond high-fidelity digitization and a unified JSON schema with traceable provenance, RoMathExam is enriched with curriculum-aligned topic tags and dense text embeddings, enabling variant detection, deduplication, and similarity-based retrieval. To overcome the lack of historical psychometric data, we propose and validate a solution complexity metric as a scalable intrinsic proxy for difficulty. Our evaluation across three frontier reasoning models (GPT-5-mini, DeepSeek-R1, and Qwen3-235B-Thinking) reveals high cross-model synchronization ($r > 0.72$), confirming the metric’s ability to isolate intrinsic mathematical depth from stochastic generation noise. We demonstrate the dataset’s utility through a longitudinal analysis that quantifies a “regime shift” from volatile historical formats to a standardized, algebra-dominant modern curriculum. RoMathExam provides a foundation for reproducible research in difficulty modeling, curriculum analytics, and LLM evaluation in low-resource linguistic contexts.

Keywords: AI in Education · Assessment Datasets · Mathematics Education · Exam Analytics · Topic Tagging · Difficulty Modeling.

1 Introduction

The rapid advancement of Large Language Models (LLMs) in mathematical reasoning has exposed a critical vulnerability in evaluation frameworks: the reliance on static, potentially contaminated benchmarks. To truly measure a model’s ability to generalize, researchers must look toward authentic, curriculum-grounded assessment data that mirrors the longitudinal and structured nature of human education.

However, for many national contexts, high-stakes examination data remains trapped in heterogeneous, unstructured formats across decades of shifting curricula. The Romanian educational landscape is a prime example, characterized by a rigorous national examination system and significant structural variations across academic tracks [13]. While recent efforts have improved digital access, a unified, longitudinal repository that enables cross-era comparison remains a critical gap.

We present RoMathExam, a longitudinal dataset of Romanian high-school mathematics examinations designed to support research on assessment analytics, curriculum evolution, and educational pipelines for tagging-based retrieval. The dataset contains 10,592 distinct problems organized into 600+ complete exam sets across multiple tracks (M1–M4), spanning 1895–2025 with a robust seven-decade core (1957–2025). We unify official exam sessions and ministry-published training variants under a single schema, enabling analyses both within a standardized era and across longer historical regimes.

Beyond OCRization and normalization, RoMathExam is enriched with curriculum-aligned topic tags and embeddings based similarity signals. Furthermore, to overcome the lack of historical psychometric data, we propose and validate a token-based solution complexity metric as a scalable intrinsic proxy for difficulty. In summary, our contributions are:

- **Longitudinal Dataset:** A high-resolution corpus of 10,592 Romanian mathematics problems spanning 130 years (1895–2025) with traceable provenance.
- **Curriculum Alignment:** A taxonomy of 70 curriculum-aligned topic tags and validation analyses enabling fine-grained trend studies.
- **Similarity Layer:** An embedding-based framework supporting deduplication, variant identification, and retrieval-based educational applications.
- **Complexity Proxy:** Quantified complexity according to the output token count of a Chain-of-thought (COT) LLM, with fluctuations validated by two other CoT LLMs.

2 Background and Related Work

2.1 Large-Scale Benchmarks and National Exam Datasets

The evaluation of mathematical reasoning in Large Language Models (LLMs) has traditionally relied on static benchmarks such as GSM8K [7] and MATH [12].

However, recent work suggests that headline scores can be fragile and susceptible to dataset contamination, which requires more robust evaluation protocols [6]. [32] and the introduction of the GSM-Symbolic benchmark by [20] further demonstrate that model performance is often sensitive to minor variations in problem phrasing or numerical constants.

A complementary direction is to evaluate models on authentic, curriculum-grounded assessment items drawn from high-stakes national examinations. Prior work has leveraged Brazil’s ENEM [17,28] to model item properties directly from question text, and the Chinese Gaokao [33] has been used as a large-scale, multi-subject benchmark for LLM evaluation. Despite these advances, structured exam repositories remain scarce for many lower-resource languages and educational systems [1,10].

In the specific context of Romanian resources, [8] recently introduced a dataset focused on standardizing contemporary mathematics problems for model evaluation. Complementing this effort, the corpus presented here prioritizes longitudinal continuity and metadata resolution to support curriculum-aware analytics. Rather than broad categorical labels, content is organized using a detailed taxonomy of 70 topic tags and precise provenance metadata—including year, session, and difficulty tracks (M1–M4). This structural depth, spanning from 1895 to 2025, is designed to facilitate historical trend analysis and fine-grained difficulty modeling alongside standard performance benchmarking.

2.2 Representation of Learning Domains

A core task in educational data mining is the association of assessment items with Knowledge Components (KCs) [14]. While recent advancements focus on automated mapping via LLMs [21,31], a persistent challenge remains in identifying "isomorphic" problems—items sharing underlying solution structures despite surface variations [16,29]. While specialized benchmarks like AlgoSimBench [15] identify these via solution-step traces, this approach is often inapplicable to historical archives lacking step-by-step gold standards. RoMathExam addresses this by providing curriculum-aligned tags that allow for the study of semantic similarity in relation to textual embeddings. This bridges the gap between surface-level Natural Language Processing (NLP) features and deep conceptual isomorphism, a limitation noted in prior cross-lingual studies [1,27].

2.3 Item Complexity and Difficulty Estimation

In the absence of student response logs, item complexity serves as a measurable proxy for difficulty. [22] and [3] define complexity as an inherent structural property—governed by solution length and conceptual density—distinguished from difficulty, which is an observed performance measure. Recent studies confirm that LLM-based features, including token counts and solution steps, correlate strongly with human-rated difficulty parameters [17,23]. Following this framework, RoMathExam utilizes token-based solution complexity to estimate relative difficulty across 130 years of exams.

To illustrate the variation in token footprint across different model architectures for an identical baseline problem, Table 1 presents the raw generation outputs for a simple arithmetic request.

Table 1. Solution generation across evaluated LLMs for the problem: “Calculate the arithmetic mean of the numbers 3, 7, 11.”

DeepSeek-R1-0528 (with Chain-of-Thought) <i><think> Am de calculat media aritmetică a numerelor 3, 7 și 11. Media aritmetică se calculează prin suma numerelor împărțită la numărul lor. Aici sunt trei numere. Dec, mai întâi adun numerele: $3 + 7 + 11 = 21$. Acum, împart suma la numărul de numere, care este 3. Deci, $21/3 = 7$. Media aritmetică ar trebui să fie 7. Să verific dacă am făcut bine. Numerele sunt 3, 7, 11. 7 este chiar la mijloc, deci media aritmetică ar trebui să fie 7, da. Formula generală este: media = (suma tuturor termenilor) / (numărul de termeni). Suma $3 + 7 + 11 = 21$, $21/3 = 7$. Cred că este corect. </think></i> Media aritmetică a numerelor 3, 7 și 11 se calculează folosind formula: Media = (Suma numerelor) / (Numărul de numere). Pași de calcul: 1. Suma numerelor: $3 + 7 + 11 = 21$. 2. Numărul de numere: 3. 3. Media: $21/3 = 7$. Rezultat: Media aritmetică este 7.
Qwen3-235B-Thinking Pentru a calcula media aritmetică a numerelor 3, 7 și 11, urmăm pașii următori: Adunăm toate numerele: $3 + 7 + 11 = 21$ Numărăm câte numere sunt: Avem 3 numere: 3, 7 și 11. Calculăm media aritmetică: Media aritmetică se obține împărțind suma numerelor la numărul lor: $Media_{aritmetic} = \frac{21}{3} = 7$. Răspuns final: 7
GPT-5-mini Media aritmetică a numerelor se calculează ca suma numerelor împărțită la numărul lor. - Formula: $M = \frac{x_1 + x_2 + x_3}{3}$ - Calcul: $M = \frac{3 + 7 + 11}{3} = \frac{21}{3} = 7$ Răspuns: media aritmetică este 7.

3 Dataset

3.1 Overview and Statistics

The RoMathExam dataset comprises **10,592 distinct problems**. Regarding metadata completeness, marking schemes are available for 4,333 items, while specific session information is preserved for 8,445 items. The content is structured using a taxonomy of 70 unique topic tags.

A defining feature of this corpus is its temporal depth. While other recent contributions to the field focus primarily on the post-2007 digital era (e.g., [8]), RoMathExam includes a significant subset of 2,320 items from the pre-2007 period. This historical coverage addresses a gap in current benchmarks, enabling the analysis of long-term curriculum evolution beyond the scope of contemporary testing standards.

3.2 Data Standardization and Profile Mapping

The dataset is distributed in JSON format via the HuggingFace Hub repository `RoMathExam/RoMathExam`⁴, with the full schema and enumeration values detailed in the README. Additionally, a permanent archival version of the dataset is hosted on Zenodo [9] to ensure long-term academic reproducibility.

To facilitate longitudinal analysis across the entire timeline, we normalized historical track designations to align with the modern four-track system. Specifically, historical "Real" and intensive mathematics variants were mapped to **M1 (Mathematics-Informatics)**, while "Uman" and general science variants were aligned with **M2 (Natural Sciences)**.

3.3 Collection and Parsing Pipeline

The raw documents were sourced from established digital archives of the Romanian mathematical community, primarily the historical collections curated by Sorin Borodi [5] and the standardized modern variant repositories of Pro-Matematica [25]. To facilitate content extraction, the entire corpus underwent optical character recognition (OCR) and conversion using Mathpix, transforming the source PDF content into Markdown format with embedded $\text{\LaTeX}$ equations.

A two-stage pipeline utilizing GPT-5-mini (via OpenAI Batch API) was employed to structure this raw text. In the first stage, the document content was processed alongside its relative file path. This path information served as a contextual anchor, allowing for the inference of information often missing from historical document headers (i.e the year of the exam might not appear in the contents, but would appear in the file name). Rather than direct text extraction, the process identified unique start/end text delimiters for each exam instance. This approach also facilitated the alignment of marking schemes (*bareme*); regardless of whether solutions were interleaved within the same file or located in paired documents, the specific text boundaries for both "Subject" and "Solution" blocks were identified simultaneously.

Atomic problems were programmatically sliced and parsed into the target schema, associating exercises with grading keys into intermediate JSON objects for validation. This transition was highly cost-efficient: high-fidelity OCR (Mathpix) cost **\$52.16**, while structural parsing and 1536-D embedding generation cost **\$58.00**. This **~\$110** total demonstrates that building century-scale datasets for regional languages is feasible under limited resource constraints, a key consideration for frugal AIED research.

⁴ <https://huggingface.co/datasets/RoMathExam/RoMathExam>

3.4 Manual Verification and Quality Assurance

To ensure dataset fidelity, we established a verification pipeline involving a mathematics teacher and 20 **volunteers** from a 12th-grade cohort (aged 18–19, M1 track). Conducted as a pedagogical exercise in data literacy, students identified subtle semantic errors [12] through a three-stage workflow: (1) **Visualization**, where validators used a lightweight HTML tool to render raw JSON and L^AT_EX into human-readable formats mirroring the test setting [7]; (2) **Annotation**, utilizing structured spreadsheets to review batched items and flag discrepancies (e.g., missing constraints, malformed equations); and (3) **Correction**, where flagged entries were manually reviewed and resolved **exclusively by the authors** to ensure archival integrity.

4 Tags and Embeddings

4.1 Taxonomy and Automated Annotation

To enhance the dataset’s utility for educational research, we developed a granular categorization schema designed to capture the specific nuances of the Romanian Baccalaureate curriculum. The taxonomy was collaboratively created with an experienced high-school mathematics teacher to prioritize specific sub-domains (e.g., "*logarithmic equations*" or "*matrix inversability*") rather than broad categories (e.g., "*Algebra*"). While this granularity increases annotation complexity, it allows for more precise content analysis.

Given the dataset’s extensive temporal depth (1895–2025), manual annotation of the full corpus was unfeasible. Leveraging recent findings on the efficacy of LLMs for zero-shot data labeling [11], we employed **Gemini 3.0 Flash**. The model was provided with the pre-defined taxonomy and a strict system prompt requiring JSON output selected exclusively from the provided list.

To prevent the model from assigning technically correct but non-essential tags, such as labeling a matrix determinant problem as "*real number calculations*", we enforced a constraint to ensure tags capture the *essence* of the problem. The limit was defined as a maximum of 3 tags for single-item exercises, and $n + 2$ tags for exercises with n sub-points.

4.2 Tagging Validation Experiments

To validate the automated annotation, we implemented a multi-stage verification process. Initial automated checks ensured that every exercise was assigned at least one tag, that all tags were part of the defined tag list and that each tag was used a statistically relevant amount of times.

Subsequently, we performed a blind manual annotation of 100 randomly sampled exercises by a domain expert. We calculated similarity metrics between the expert’s tags and the LLM’s tags. The results indicate a high degree of alignment [18], as detailed in Table 2.

Qualitative analysis of the discrepancies revealed three primary disagreement types:

Table 2. Similarity metrics between Expert and LLM annotations (N=100)

Metric	Value
Jaccard Similarity	0.8847
Precision	0.8811
Recall	0.9662
F1 Score	0.9217
Cohen’s Kappa	0.9477

- **Granularity Noise:** The LLM included generic tags for minor steps (e.g., tagging trigonometric operations within a matrix problem).
- **Method-Dependent Tagging:** The LLM inferred specific solution paths (e.g., tagging "derivatives" for limits solvable by L’Hôpital’s rule).
- **Ontological Ambiguity:** Arising when multiple valid strategies existed (e.g., tagging "vectors" vs. "geometry" for shape vertices).

These experiments suggest that the automated tags are highly reliable in identifying the core content of the exercises. We argue that a perfect agreement (1.0) is practically unattainable with a taxonomy of this granularity due to the inherent subjectivity of mathematical problem classification, a well-documented phenomenon in linguistic and semantic annotation tasks [2]. While using broader categories (e.g., Geometry vs. Analysis) would almost certainly generate even higher agreement, it would drastically reduce the utility of the dataset.

4.3 Textual Embeddings

Semantic representations were generated using OpenAI’s `text-embedding-3-small` model, producing 1536-dimensional vectors for each item. To ensure that similarity metrics reflect only the mathematical content, the model was provided exclusively with the raw problem statements, strictly excluding secondary information like topic tags, difficulty proxies, and temporal metadata. However, because this is a general-purpose text embedding architecture, a notable limitation is its tendency to over-index on lexical surface features rather than underlying mathematical structures. Table 3 illustrates this behavior, underscoring the necessity of pairing semantic embeddings with explicit, curriculum-aligned topic tags to achieve true structural isomorphism in educational retrieval tasks.

4.4 Longitudinal Curriculum Analysis

With the granular tagging taxonomy validated and high-dimensional embeddings generated, we analyzed the temporal evolution of the curriculum, revealing significant pedagogical “phase transitions”.

Curriculum Consistency and Regime Shifts. To quantify exam stability over time, we computed two *Temporal Similarity Matrices* (Figure 1): one calculating the cosine similarity between annual topic distribution vectors [19], and another measuring the cosine similarity between annual embedding centroids.

Table 3. Example of embedding limitation. The `text-embedding-3-small` model computes a 95.8% cosine similarity between the original Romanian items due to near-identical text formulation, despite significant differences in underlying solution methodology. English translations are provided for clarity.

<i>Original:</i> Rezolvați în mulțimea numerelor reale ecuația $\sqrt[3]{x^2 - 1} = 2$.
<i>Translated:</i> Solve the equation $\sqrt[3]{x^2 - 1} = 2$ in the set of real numbers.
<i>Original:</i> Rezolvați în mulțimea numerelor reale ecuația $\sqrt{2x^2 - 3x + 2} = x$.
<i>Translated:</i> Solve the equation $\sqrt{2x^2 - 3x + 2} = x$ in the set of real numbers.

Both heatmaps independently reveal a striking antithesis between the 20th and 21st centuries. Pre-2000 exams exhibit low cross-temporal similarity, indicating a highly volatile curriculum. In contrast, the post-2000 era displays a highly stable, standardized framework in both explicit tag-space and latent embedding-space. This high congruence validates that the observed “regime shift” is a genuine property of the curriculum’s evolution, simultaneously confirming the embeddings’ ability to capture pedagogical essence.

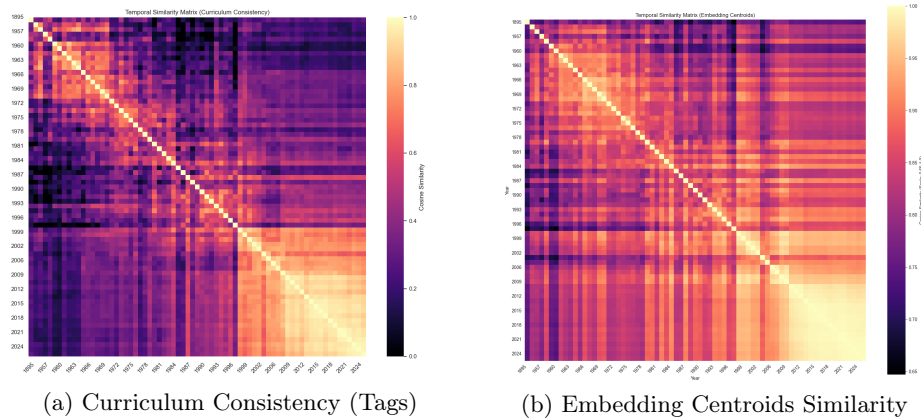


Fig. 1. Temporal Similarity Matrices comparing year-to-year consistency.

Topic Distribution Drift. Furthermore, an analysis of major domains confirms a long-term concept drift (Figure 2). The early 20th-century emphasis on Geometry and Analysis has been superseded in the modern era by Algebra, which now consistently accounts for $\approx 60\%$ of the exam content. This structural evolution emphasizes the need for temporally-aware training splits in AIED applications to prevent performance overestimation in non-stationary environments [30].

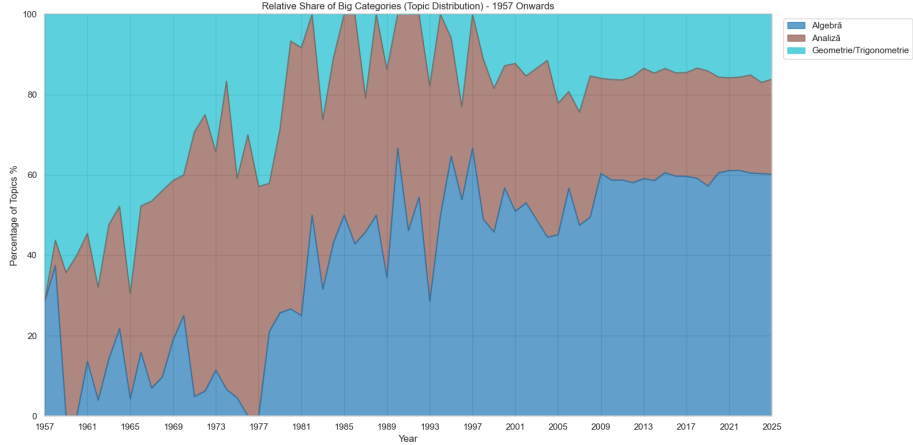


Fig. 2. Relative Share of Major Mathematical Categories (1957–2025). The plot highlights the gradual dominance of Algebra in the modern curriculum.

5 Difficulty Estimation via Token-Based Solution Complexity

5.1 Motivation: The Limitations of Traditional Difficulty Metrics

Establishing a consistent difficulty metric across a 130-year span is challenging due to the lack of historical student performance logs and the confounding nature of modern psychometrics. Traditional difficulty proxies, such as pass rates and human perception, are heavily compromised by sociopolitical variables. For instance, the introduction of rigorous anti-corruption measures (e.g., CCTV monitoring in 2011) caused a precipitous drop in pass rates entirely unrelated to mathematical complexity [4]. Concurrently, the shift toward standardized testing has altered the subjective perception of difficulty, often replacing creative reasoning with rote memorization [24].

To circumvent these sociopolitical and perceptual confounders, we propose Output Token Complexity as an objective, intrinsic proxy. By utilizing LLMs as “synthetic students”, the mathematical signal is isolated from external testing stressors. The central hypothesis is that for a sufficiently capable reasoning model, the length of the generated solution path correlates strongly with the logical depth and number of inference steps required, providing a sterile, scalable measure of complexity.

5.2 Methodology and Architecture-Agnostic Validation

To operationalize this metric, a batch processing pipeline was implemented using `gpt-5-mini` as the primary evaluator for the full dataset ($N = 10592$). To mitigate the risk of prompt-induced verbosity, a minimal system instruction

was employed: "*Ai rolul de a rezolva probleme de matematică*" ("Your role is to solve math problems"). This unconstrained prompt enables the model to dynamically determine the necessary depth of explanation based solely on the problem's requirements.

Validation was conducted to ensure that the resulting complexity signal is an intrinsic property of the dataset rather than an artifact of a specific model architecture. A cross-model consistency check was performed using two additional high-reasoning models, Qwen/Qwen3-235B-Thinking and DeepSeek-R1-0528, restricted to the "Summer" and "Simulation" sessions of the standardization era (2010–2025), as the two models are primarily used to validate the difficulty evolution in the results from gpt-5-mini.

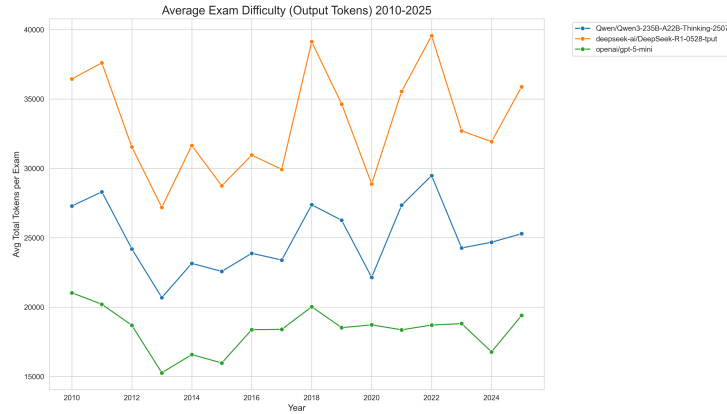


Fig. 3. Cross-model complexity trends (2010–2025). High synchronization ($r > 0.72$) confirms intrinsic difficulty spikes between the models.

The analysis, visualized in Figure 3, reveals a high synchronization between models ($r > 0.72$). Although absolute token counts vary due to tokenizer distinctiveness, the relative longitudinal fluctuations remain consistent across all three architectures. Validating this proxy via multi-model reasoning chains cost approximately **\$93.80**, confirming that large-scale difficulty estimation is computationally accessible.

5.3 Findings: Structural Evolution and Syllabus Dynamics

Applying the token-complexity metric confirms that National Simulations are consistently more difficult than final Summer sessions. Across the 2010–2025 interval, simulations yielded statistically higher token counts, supporting the hypothesis that diagnostic tests prioritize complex, integrative problem-solving to incentivize study.

Longitudinally, Figure 4 reveals a shift in the definition of difficulty. Historical exams (1950–2000) display high variance driven by idiosyncratic, high-complexity outliers. In contrast, the modern era (2010–2025) exhibits a “Low Complexity / High Consistency” profile: while mean per-item complexity has decreased, the aggregate token volume per exam has converged into a stable band. This reflects a pedagogical transition from difficulty based on creative derivation to one requiring sustained cognitive stamina across a standardized algorithmic load.

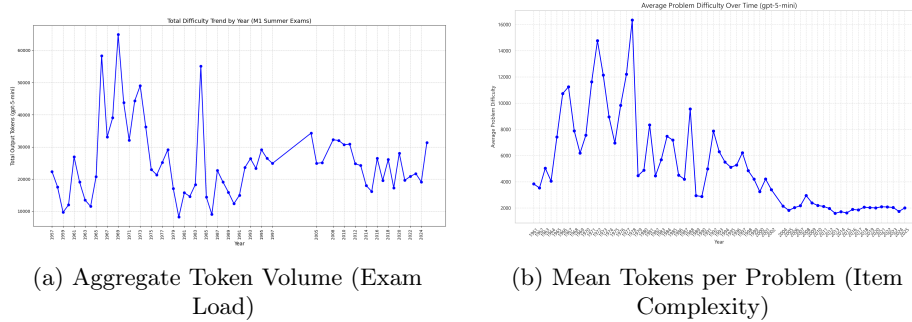


Fig. 4. The Stabilization of Difficulty Metrics. Historical exams (pre-2000) display significant stochastic variance (a), often driven by isolated, high-complexity outliers. The modern standardized era (2010–2025) demonstrates a convergence toward stability: while individual item complexity has decreased (b), the aggregate cognitive load is maintained through a higher volume of items, resulting in a consistent total output.

6 Educational Use Cases

6.1 Personalized Practice and Isomorphic Retrieval

For students, the primary challenge in exam preparation is often not the lack of material, but the inability to locate specific practice problems that target their individual weaknesses. Traditional keyword searches are often too broad. RoMathExam addresses this by enabling *embedding-based retrieval*, allowing students to perform “isomorphic practice.”

By utilizing the semantic similarity layer (Section 4.3), a student can input a specific problem they found difficult and instantly retrieve the k -nearest neighbors from the historical archive. As shown in Table 4, this process identifies problems that share the same underlying logical structure or solution strategy, even if the numerical values or surface phrasing differ. This allows for targeted reinforcement of specific algorithmic techniques (e.g., irrational equations involving radicals) rather than generic topic review. Furthermore, because the modern examination era is characterized by a highly standardized blueprint (as established in Section 4.4), semantic similarity acts as a strong proxy for difficulty

alignment. Consequently, retrieved variants are not only topically related but typically demand a comparable level of cognitive complexity.

Table 4. Examples of Isomorphic Retrieval. The system identifies historical variants that share the same underlying logic as the query problem, enabling targeted student practice. English translations of the problem statements are provided for clarity.

Query Problem (Student Input)	Retrieved Isomorphic Variant
<i>2025, Special Session, M2, Original:</i> Rezolvați în mulțimea numerelor reale ecuația $\sqrt{2-x} = \sqrt{x^2+x-1}$. <i>Translated:</i> Solve the equation $\sqrt{2-x} = \sqrt{x^2+x-1}$ in the set of real numbers.	<i>2011, Model, M2, Original:</i> Rezolvați în mulțimea numerelor reale ecuația $2 - \sqrt{2-x} = x$. <i>Translated:</i> Solve the equation $2 - \sqrt{2-x} = x$ in the set of real numbers.
<i>2023, Special Session, M2, Original:</i> În reperul cartezian xOy se consideră punctele $A(1, -5)$ și $B(5, 5)$. Determinați distanța de la punctul O la mijlocul segmentului AB. <i>Translated:</i> In the Cartesian coordinate system xOy, consider the points $A(1, -5)$ and $B(5, 5)$. Determine the distance from point O to the midpoint of the segment AB.	<i>2010, Model, M2, Original:</i> În reperul cartezian xOy se consideră punctele $A(-1, -2)$, $B(1, 2)$ și $C(2, -1)$. Calculați distanța de la punctul C la mijlocul segmentului AB. <i>Translated:</i> In the Cartesian coordinate system xOy, consider the points $A(-1, -2)$, $B(1, 2)$ and $C(2, -1)$. Calculate the distance from point C to the midpoint of the segment AB.

6.2 Stylistic Alignment for Automated Item Generation

A significant challenge in utilizing LLMs for automated item generation is the "linguistic alignment gap": generic models often produce mathematically correct but stylistically unnatural phrasings that deviate from established national standards. RoMathExam serves as a high-fidelity reference corpus for *Retrieval-Augmented Stylistic Alignment*. By retrieving existing problem statements that are structurally isomorphic to a draft input (e.g., a raw equation or abstract concept), an LLM can be prompted to contextualize abstract mathematical structures within the canonical linguistic register of the Romanian curriculum. This process ensures that generated items adhere to the specific syntactic conventions and imperative formulations (e.g., "Se consideră...", "Să se demonstreze că...") characteristic of the historical archive, thereby automating the production of exam-ready content [26].

7 Ethics and Limitations

Provenance. ROMATHEXAM is compiled from official Romanian high-school mathematics exam materials and ministry-published training variants released for public use.

Copyright and Licensing. The dataset artifacts—including the unified JSON schema, curriculum-aligned topic tags, derived complexity signals, embeddings, and associated processing code—are released under the **MIT License**. We do not claim copyright over the original mathematical problem statements, which remain the property of their respective administrative publishers. Under the MIT License, users are free to use, modify, distribute, and commercially exploit the structured dataset and code, provided the original copyright notice and permission notice are included.

Privacy and Responsible Use. The dataset contains no personal identifying information. We position RoMathExam for research and formative learning and strongly discourage its use for real-time assistance or cheating in proctored educational settings.

Derived Difficulty Proxy. Token-based complexity is an intrinsic, protocol-dependent proxy. It is not a human psychometric difficulty estimate and may vary based on model architecture and generation settings.

8 Conclusion and Future Work

Conclusion. In this work, we introduced RoMathExam, a longitudinal dataset of Romanian high-school mathematics exams spanning 1895–2025. By providing a unified JSON schema, curriculum-aligned topic tags, and a cross-LLM validated token-based complexity proxy, we address the scarcity of structured, long-term assessment data for regional languages. Our analysis of regime shifts in the curriculum highlights the dataset’s utility for understanding the evolution of national pedagogical standards.

Future Work. To extend this research, we propose three main directions: (1) augmenting the complexity metric with sub-question dependency ratios and step-to-answer estimates to capture structural complexity; (2) developing machine-readable diagram representations for geometry-heavy Olympiad subsets to enable multimodal expansion; and (3) releasing an English-translated version to facilitate global cross-national pedagogical benchmarking.

Reproducibility and Data Access. To support the AIED community, we release the full RoMathExam corpus (including metadata and 70-topic tags), precomputed 1536-D semantic vectors to minimize community costs, and the complete set of Python scripts used for parsing, complexity calculation, and synchronization.

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